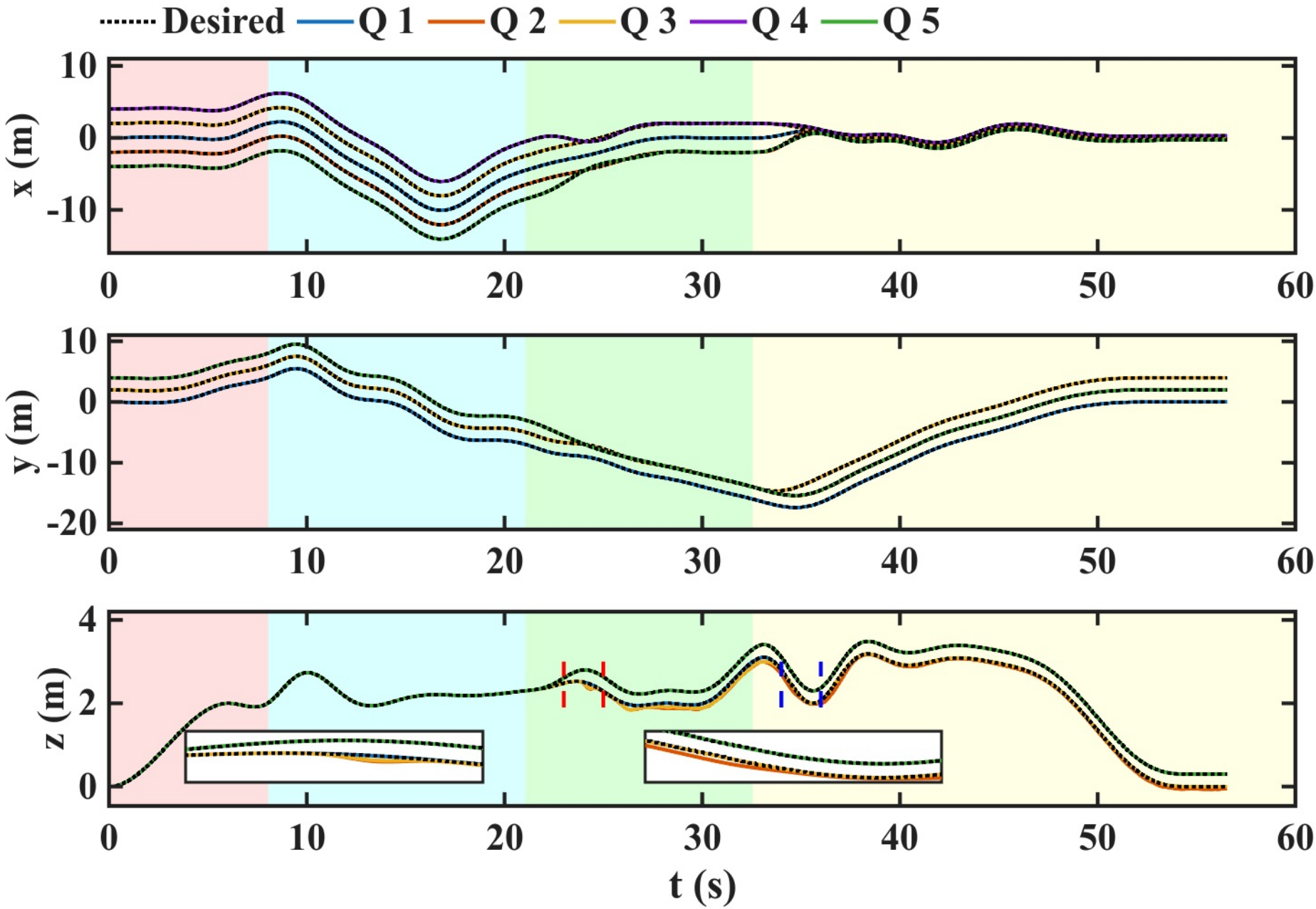
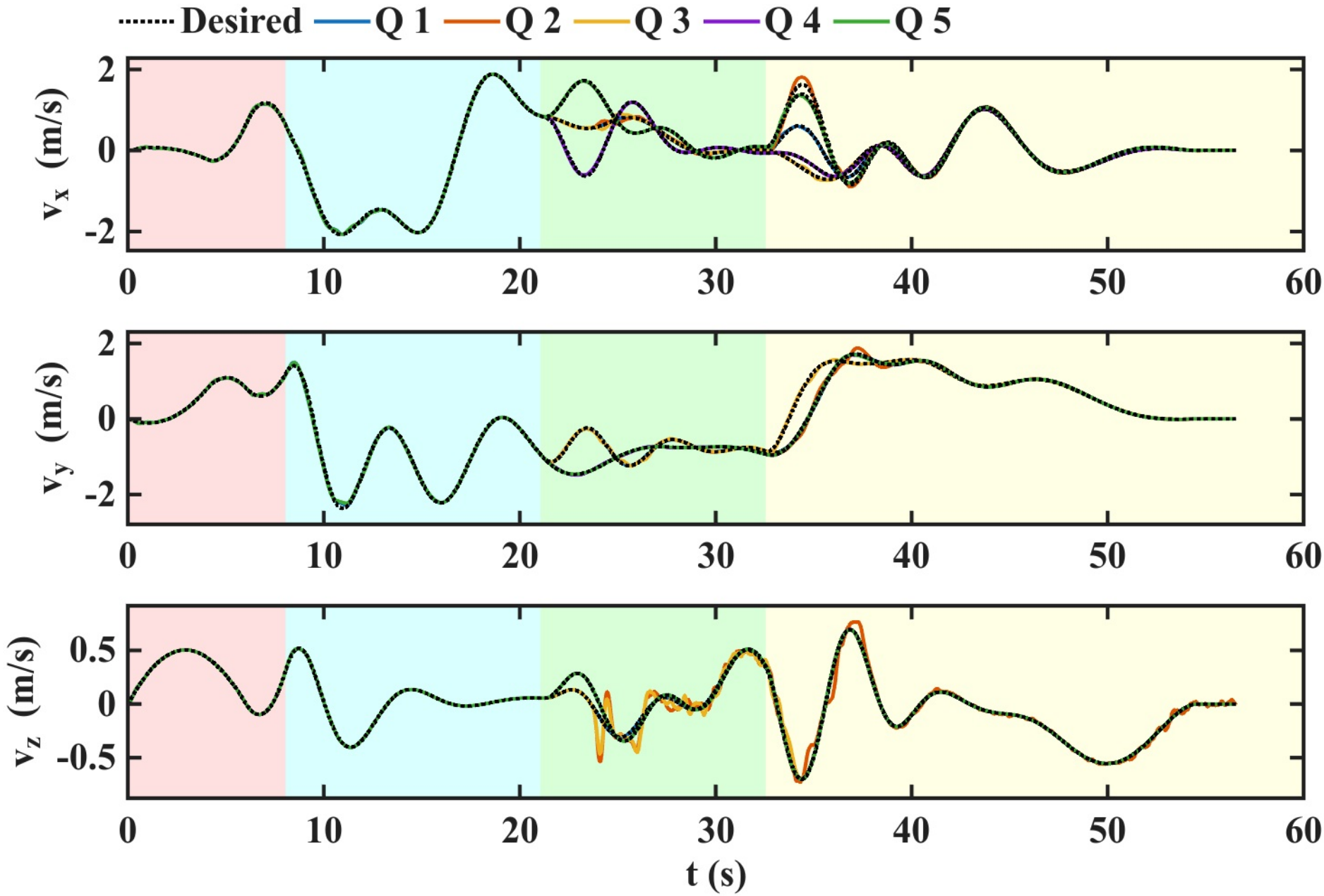
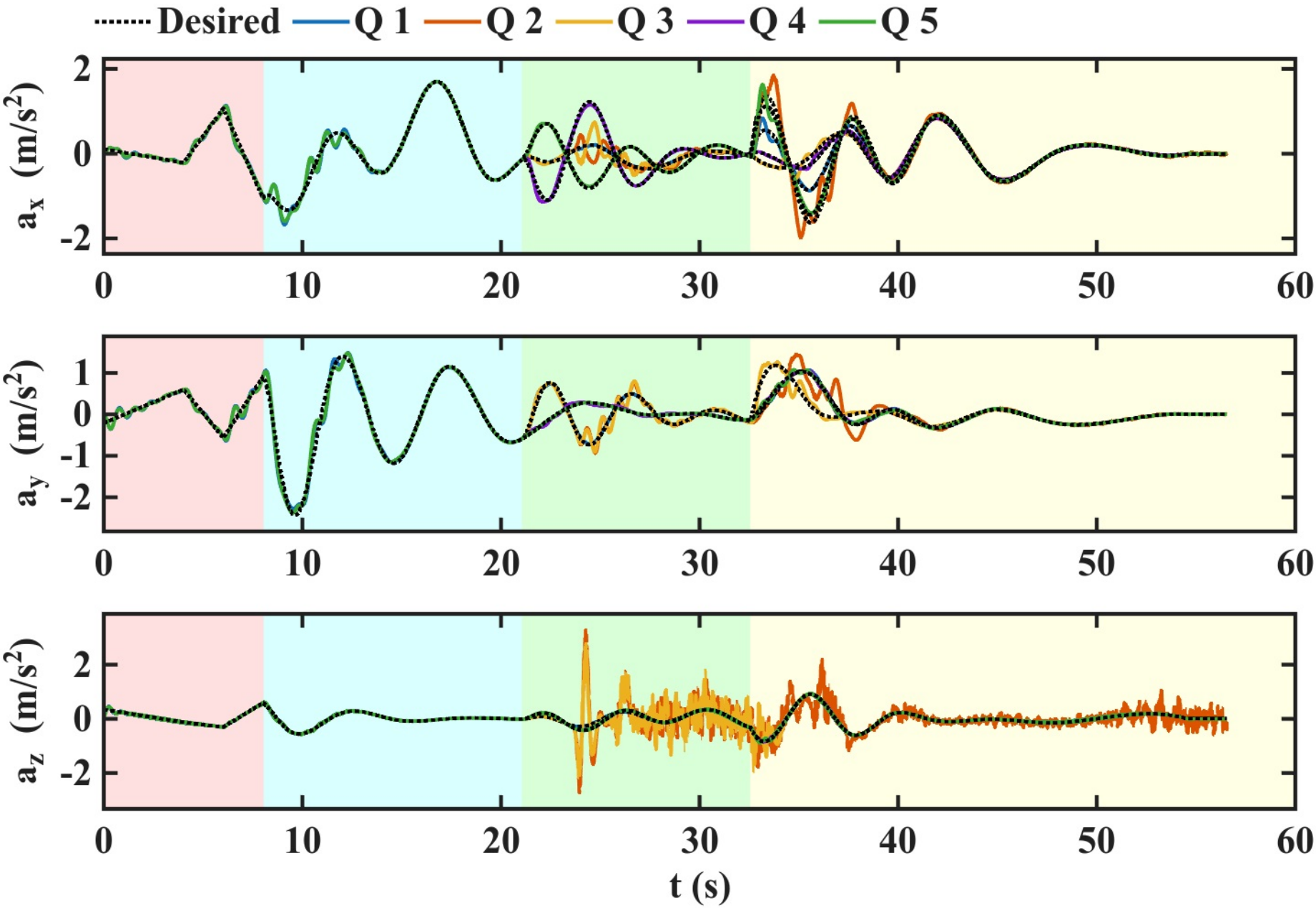
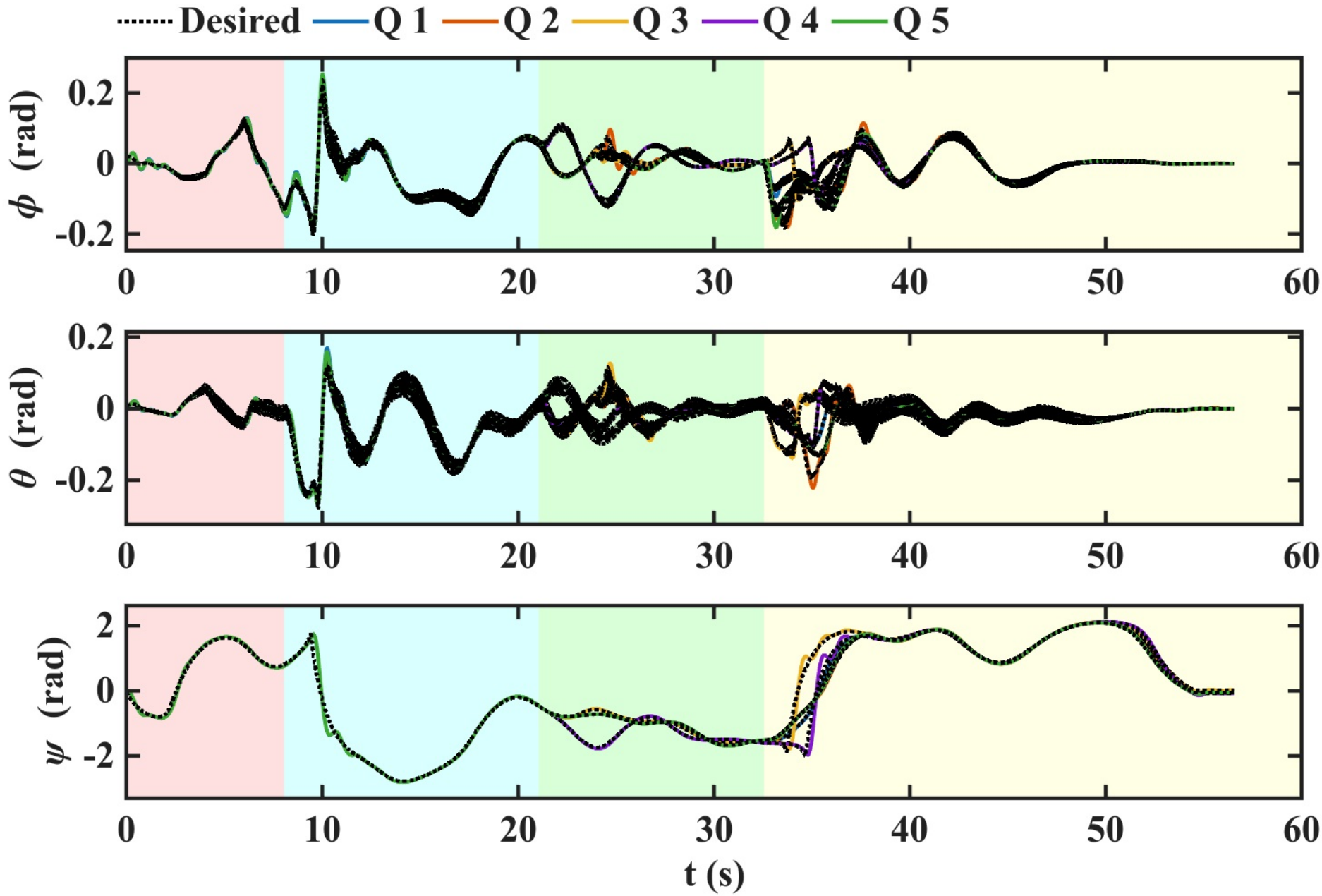
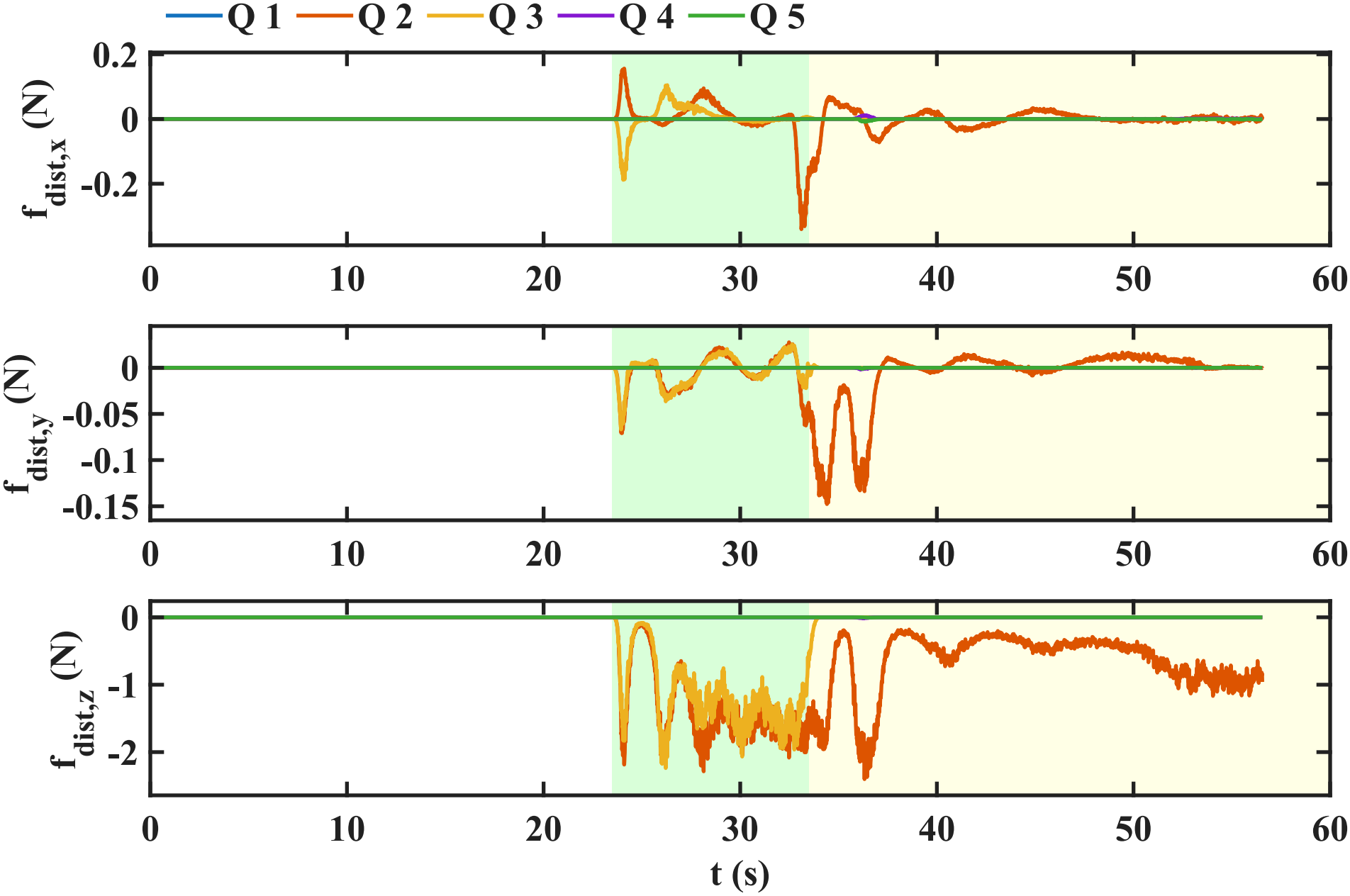
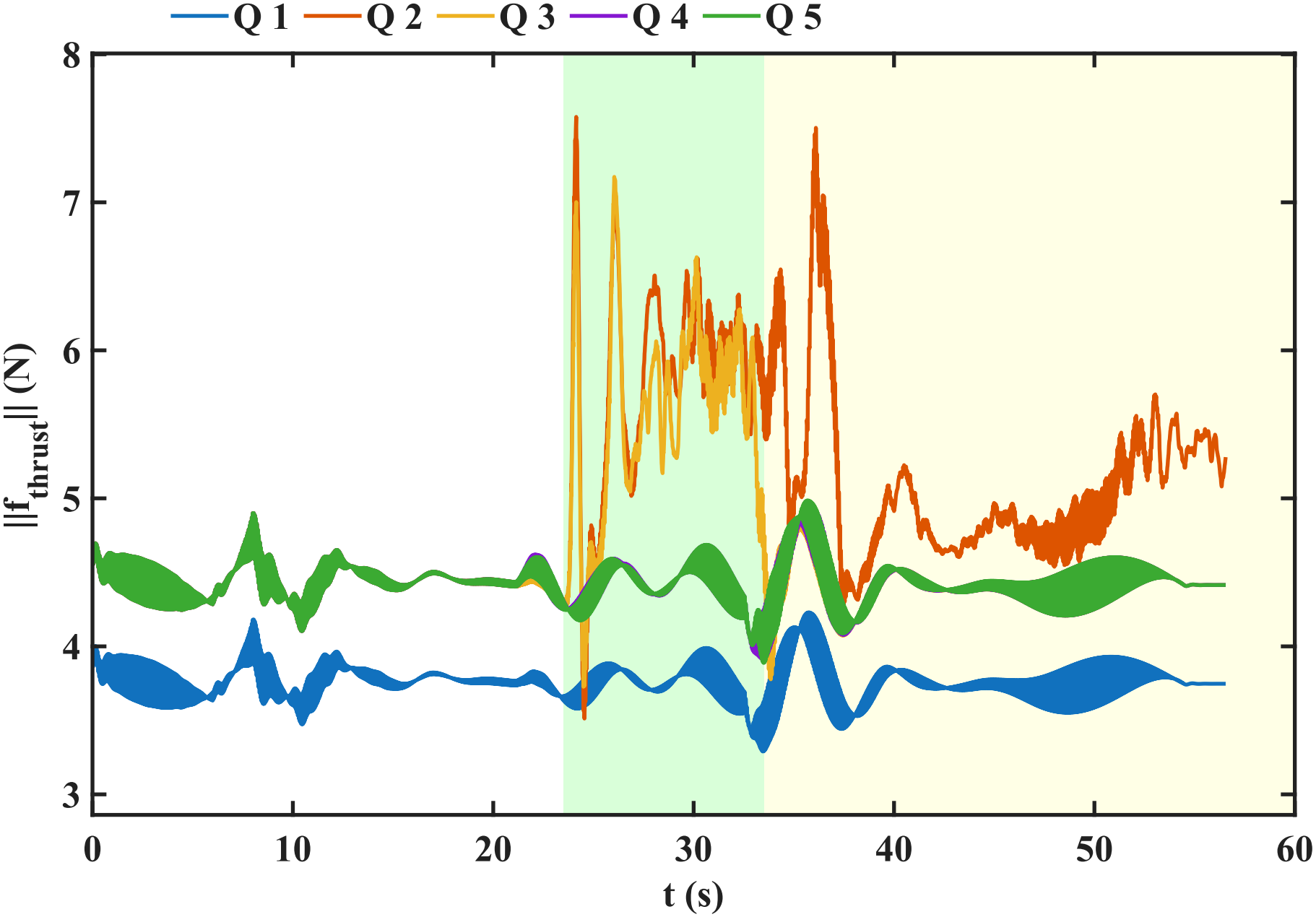

Desired
Q 1
Q 2
Q 3
Q 4
Q 5
x (m)
y (m)
z (m)
t (s)
10
0
-10
-20
4
2
0
10
20
30
40
50
60

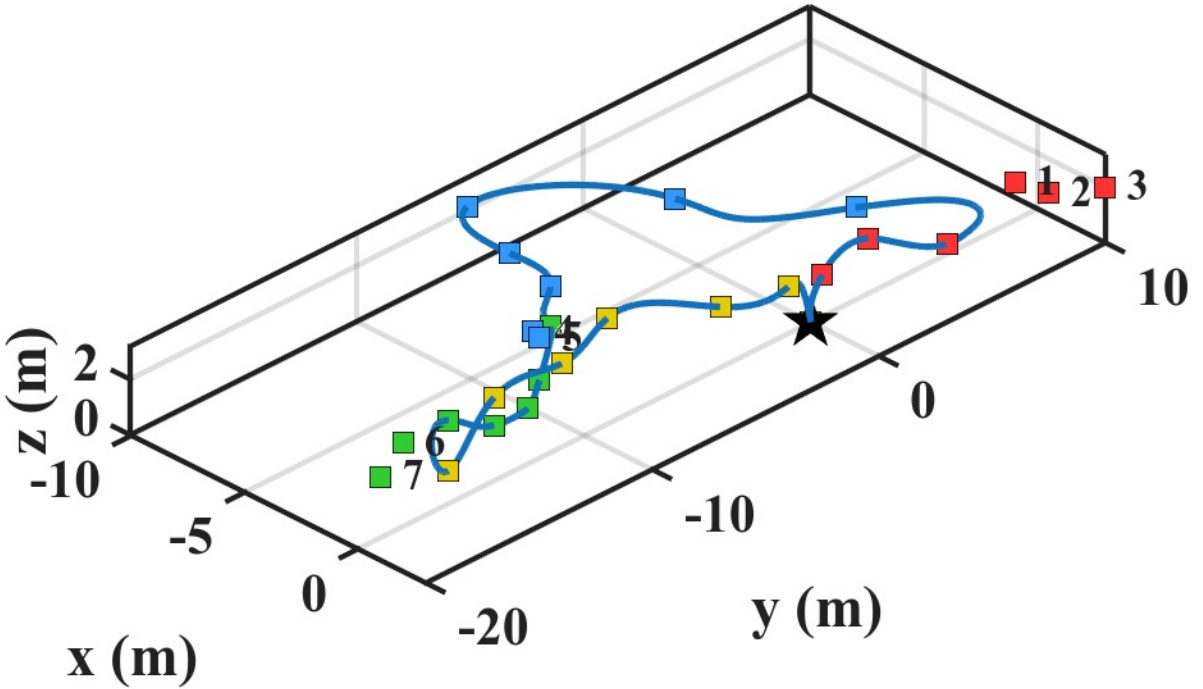

Desired
Q 1
Q 2
Q 3
Q 4
Q 5
v_x (m/s)
v_y (m/s)
v_z (m/s)
2
0
-2
0.5
-0.5
0
10
20
30
40
50
60
t (s)

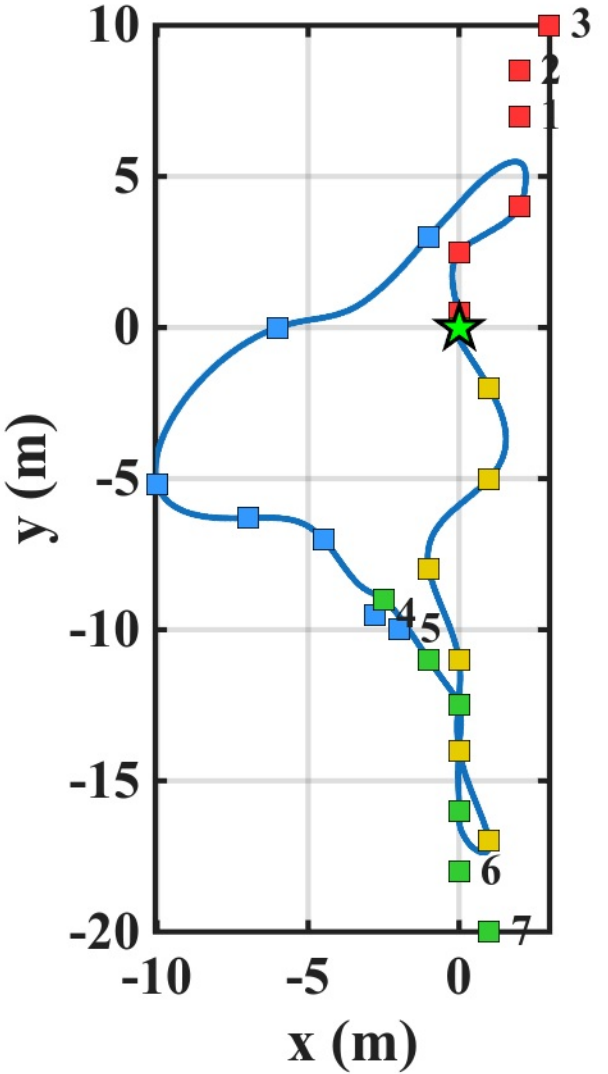

Desired
Q 1
Q 2
Q 3
Q 4
Q 5
a_x (m/s^2)
a_y (m/s^2)
a_z (m/s^2)
t (s)

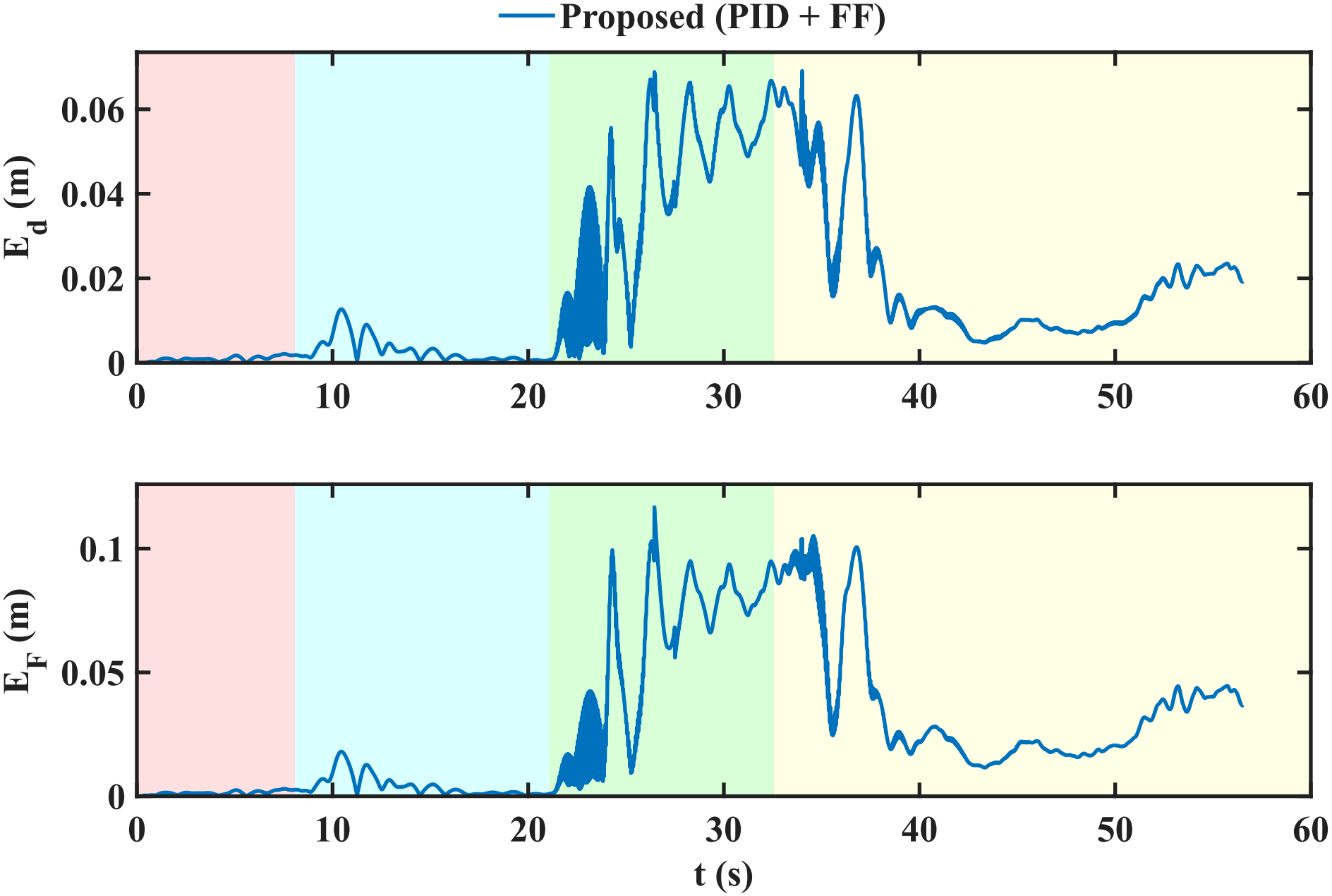

Desired
Q 1
Q 2
Q 3
Q 4
Q 5
ϕ (rad)
θ (rad)
ψ (rad)
0.2
0
-0.2
2
-2
0
10
20
30
40
50
60
t (s)


Q 1
Q 2
Q 3
Q 4
Q 5
0.2
0
-0.2
$f_{dist,x}$ (N)
0
10
20
30
40
50
60
0
-0.05
-0.1
-0.15
$f_{dist,y}$ (N)
0
10
20
30
40
50
60
0
-1
-2
$f_{dist,z}$ (N)
0
10
20
30
40
50
60
t (s)


Q 1
Q 2
Q 3
Q 4
Q 5
8
7
6
5
4
3
$||f_{thrust}||$ (N)
0
10
20
30
40
50
60
t (s)


z (m)
2
0
-10
-5
0
x (m)
-20
-10
0
10
y (m)
1
2
3
4
5
6
7


y (m)
10
5
0
-5
-10
-15
-20
-10
-5
0
x (m)
1
2
3
4
5
6
7


Proposed (PID + FF)
E_d (m)
0.06
0.04
0.02
0
0
10
20
30
40
50
60
E_F (m)
0.1
0.05
0
0
10
20
30
40
50
60
t (s)